SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

Weightage: 10%

QUESTIONS: **Total Marks: 20**

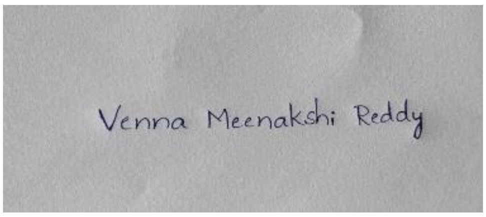
1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.

2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I **VENNA MEENAKSHI REDDY(192471048)** *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding
Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

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CO4 - AT3
Class Test

Q,1) Fundamental concepts used in web development

1. XML (extensible Markup Language)

XML is a markup language used to store, organize, and transfer data.

→ unlike HTML, XML doesnot have predefined tags. Users can create their own tags.

Purpose:-

→ Transfers data between different applications.

→ Stores structured data.

Role:- XML acts as a data storage & data exchange format.

2. JSP (Javaserver Pages)

Java Server Pages (JSP) is a Java-based server-side technology used to create dynamic web pages.

Purpose :-

To generate web pages dynamically based on user requests and data.

Role:-

JSP acts mainly as the presentation layer, displaying dynamic data to users.

3. MVC Architecture :- (Model View Controller).

MVC is a software architecture that divides an application into model, view, and controller.

Purpose:-

To seperate data, user interface, and application control logic.

Role:-

MVC organizes the application into separate components, making it easier to develop, maintain, and modify.

Model : Handles data and business logic.

View : Displays information to the user.

controller : Receives requests and controls the application flow.

4. XSLT - extensible style sheet Languge Transformations

→ XSLT is a language used to transform XML documents into other formats such as HTML.

Purpose:

To convert and format XML data.

Role:-

XSLT acts as a transformation tool.

5. XML Schema

XML Schema is used to define the structure, elements, attributes, and data types of an XML document.

Purpose:

To specify rules that an XML document must follow.

Role:

XML Schema acts as a validation mechanism.

6. JSTL - JavaServer Pages Standard Tag Library

JSTL is a collection of predefined tags used in JSP pages.

Purpose:-

To perform common operations such as looping, conditions, and displaying data.

Role:-

JSTL helps in the presentation layer of JSP.

7) PHP - Hypertext preprocesser

PHP is a server-side scripting language used to develop dynamic web applications.

Purpose:- To process requests and generate dynamic web content.

Role:- PHP acts as a server-side processing Technology.

8) DataBase Connectivity

DataBase Connectivity is the process of Connecting a webapplication to a database.

Purpose:-

To store, retreive, update, and delete application data.

Role:-

It provides the Communication link between the webapplication and database.

9. Parser:

A Parser is a software component that reads and analyses structured data according to its syntax.

Purpose:-

To check and interpret data so that an application Can process it.

Role:-

A Parser reads and extracts information from dataformats.

Q,2) Interconnection of these technologies in a web application:-

All these technologies can work together to build a Complete web application.

Step 1:- User sends a request.

A user accesses a web application through a browser and submits information using a form.

User
↓
Webbrowser
↓
Server.

Step 2:- JSP (or) PHP Performs Server-Side processing.

The request is processed by server-side technologies Such as JSP/PHP.

User request
↓
JSP | PHP

Step 3:- MVC organizes the Application.

MVC seperates the application into parts:

User

↓

Controller

↓

model

↓

Database

↓

Controller

↓

View (JSP)

↓

User

Role of each Component:-

→ model : Handles data and database operations.

→ View : Displays the output to the user using JSP.

→ controller : Receives requests and Controls the flow of the application.

Step 4:- Database Connectivity :-

The model connects to the database using database connectivity.

Model

↓

Database connectivity

↓

MySQL Database

The application Can retreive information Such as:

→ Student name

→ Registration Number

→ Course Details.

Step 5:- XML Stores and tramsfers data.

XML Can be used to exchange structured information between different applications.

Step 6:- XML Schema validates XML.

XML Document

↓

XML Schema Validation

↓

Valid XML.

Step 7:- Parser processes XML Data

XML Document

↓

XML parser

↓

Extracted data

↓

Application

Step 8:- XSLT Transforms XML.

XML Data + XSLT

↓

HTML Page → Browser.

Step9:- JSTL helps JSP Display data.

Database

↓

Model

↓

Jsp+JSTL

↓

web Page.

when JSP is used as the view, JSTL helps display database data without writing much Javacode.

Overall Architecture

User

↓

webbrowser

↓

Controller (MVC)

↓ Model → Database → XMLData → XML Schema → Parser → XSLT → HTMLoutput

↓ View JSP → JSTL/HTML